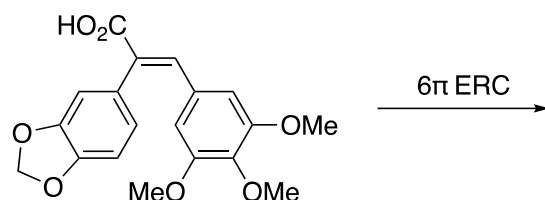


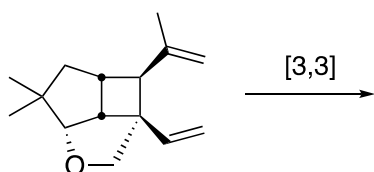
Pericyclic Reactions and the Diels-Alder Reaction

- In a large-scale synthesis of the anticancer drug steganone, electrocyclic ring closing was used strategically.

- Provide the product of 6π electrocyclic ring closing of the compound below and use curved arrows to depict the mechanism.

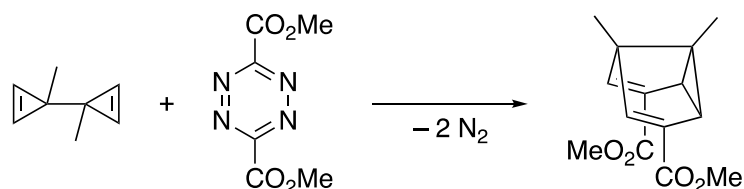


- Upon exposure to air and iodine, the product of ring closing loses the equivalent of a molecule of H_2 . Draw the product of this oxidation process.
- The Cope rearrangement is a [3,3]-sigmatropic rearrangement of 1,5-dienes with great importance in organic synthesis. The reaction requires a driving force to proceed to a synthetically useful extent at equilibrium. Draw the product of the Cope rearrangement below, use curved arrows to depict the mechanism, and explain why the product is more stable than the starting material.

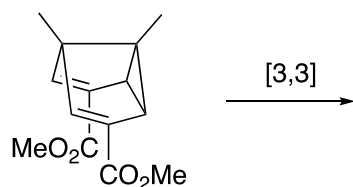


- [Bullvalenes](#) are a fascinating class of fluxional molecules, which engage in rapid and degenerate sigmatropic rearrangements. The result is electron delocalization over single bonds!

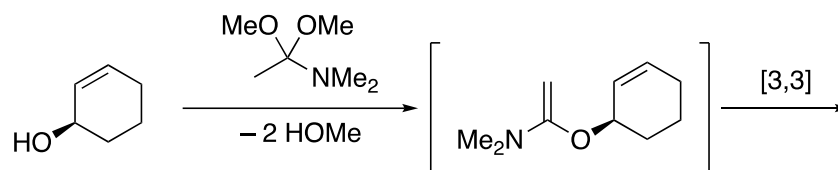
- Provide a mechanism for the reaction below. All elementary steps are pericyclic in nature—in fact, all the elementary steps are either $(4 + 2)$ or retro- $(4 + 2)$!



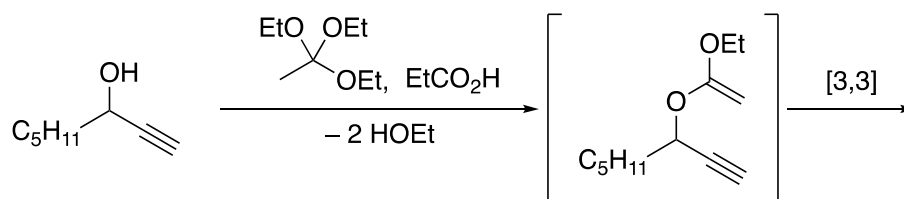
- The product above engages in a rapid Cope rearrangement. Use curved arrows to depict electron flow for this rearrangement.



- C. At room temperature, considering your answer to the previous problem, how many distinct signals would you expect in the ^1H NMR spectrum of this compound?
4. The Claisen rearrangement is likewise a [3,3]-sigmatropic rearrangement, but in this reaction the 1,5-diene contains an oxygen atom.
- A. Draw the product of Claisen rearrangement of the intermediate in brackets below and use curved arrows to depict the mechanism.



- B. Unlike the Cope rearrangement, the Claisen rearrangement typically involves an inherent driving force. Explain why Claisen rearrangements are thermodynamically favorable in general.
- C. The Claisen rearrangement even works in substrates containing *triple* bonds. Provide the product of the reaction below and use curved arrows to depict the mechanism.

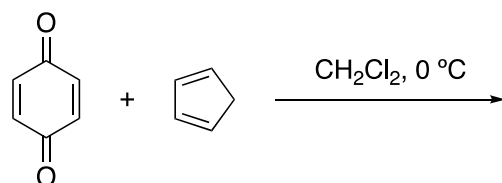


- D. The transition state of the Claisen rearrangement resembles a cyclohexane chair. Provide a structure for this transition state that accounts for the stereochemical outcome below.

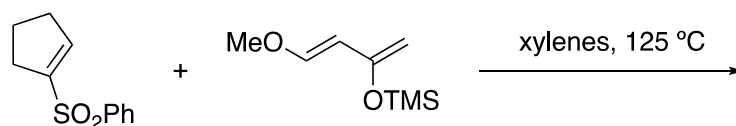


5. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

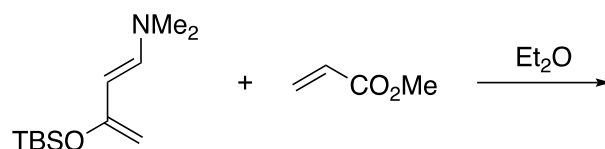
A.



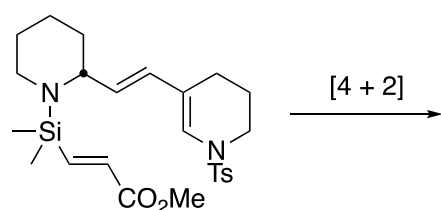
B.



C.

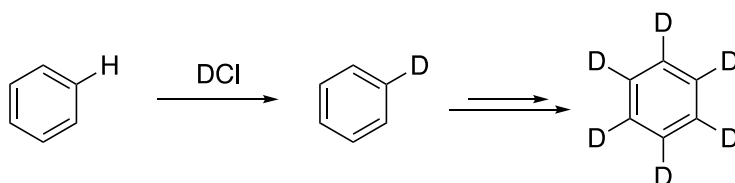


D.



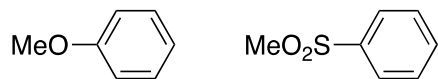
Aromatic Substitution Reactions

6. Deuterium (²H, D) is an isotope of hydrogen. When DCl gas is bubbled through a solution of benzene, the peak in its ¹H NMR spectrum at 7.34 ppm slowly disappears.

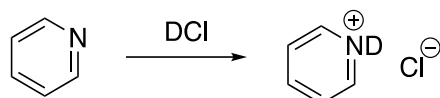


- A. Draw a mechanism for the reaction above (just incorporate one deuterium atom) that explains this observation. The mechanism should include one reactive intermediate; include all resonance forms of this intermediate in your response.
- B. When the substituted benzenes below are subjected to similar conditions, deuterium incorporation is much faster at some positions than others. Indicate the product of

treatment with DCl gas in each case, assuming the reaction is stopped after the relatively fast substitutions are complete.

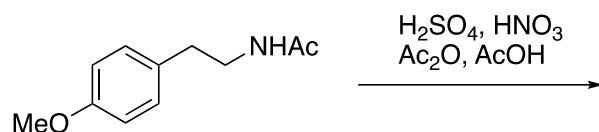


- C. When DCl gas is bubbled through pyridine, a product with the formula C_5H_5DNCl is isolated. There are three possible structures for this product. One is the ionic salt below; draw the other two.

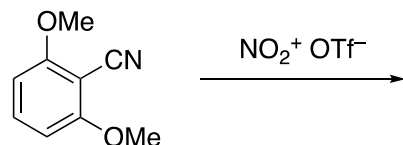


7. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

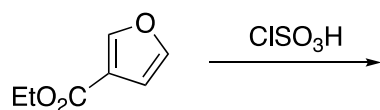
A.



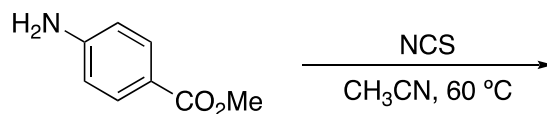
B.



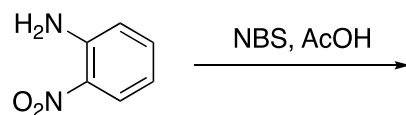
C.



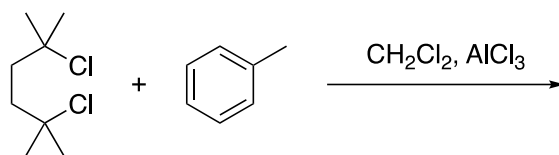
D.



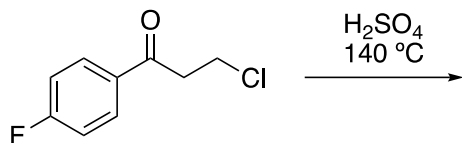
E.



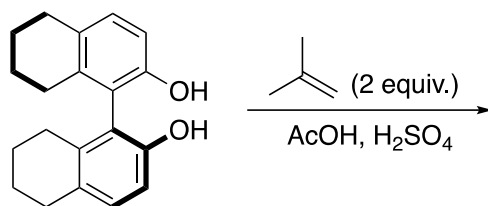
F.



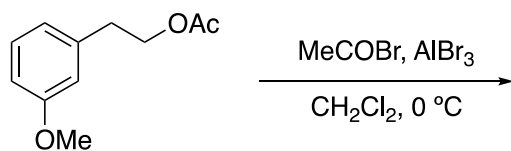
G.



H.

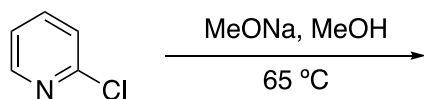


I.

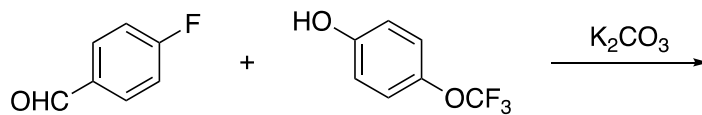


8. Provide the neutral organic product of each reaction below after appropriate workup. If you expect a reaction to yield multiple isomers, draw only the major product.

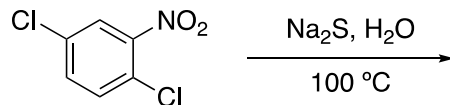
A.



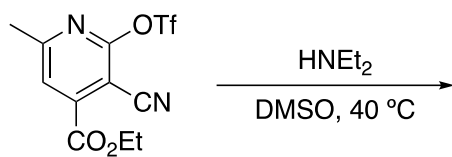
B.



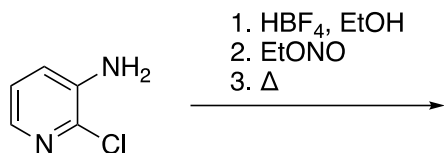
C.



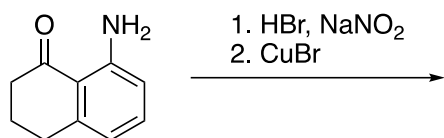
D.



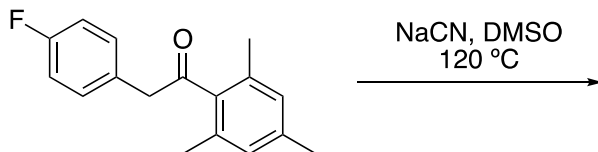
E.



F.



G.



H.

